

1 Abstract

This report details the quantitative evaluation of chromatographic peak characteristics derived from a filtered subset of the ‘chromatography_combined.csv’ dataset. The primary objective was to assess peak shape quality and detect anomalies indicative of process instability or measurement errors. Data was restricted to records containing non-null precise volume boundaries, representing approximately 88% of the total dataset, thereby introducing a selection bias that excludes flat baseline fractions. To mitigate data quality issues, negative cumulative volume offsets were normalized to zero, and peak width was calculated as the difference between maximum and minimum precise volumes. The analysis employed robust statistical methods, including the Interquartile Range (IQR) and Z-scores (threshold ≥ 3), to identify outliers. Visualizations confirmed the distribution of peak widths across experimental runs, revealing numerous outliers in specific runs that suggest potential process deviations. While the methodology for outlier detection and peak shape assessment was methodologically sound, the absence of raw data snippets and specific statistical thresholds in the visualizations limits a comprehensive evaluation of outlier significance. Future work should incorporate raw data tables and explicit statistical criteria to fully characterize the impact of selection bias on overall run stability.

2 Introduction

In biopharmaceutical manufacturing, the integrity of chromatographic peaks serves as a critical indicator of product quality and process stability. This analysis focuses on the quantitative evaluation of peak characteristics—specifically width, height, and symmetry—to identify deviations that may signal column degradation, sample impurities, or process instability. The dataset under review, ‘chromatography_combined.csv’, contains over 1 million long-form time-series records. However, peak shape metrics are only available for fractions containing detectable peaks, resulting in a significant selection bias where flat baseline data is excluded. This report outlines the motivation to evaluate peak integrity under these constrained conditions, detailing the methodology used to normalize volume offsets, calculate peak metrics, and detect outliers using statistical thresholds. The ultimate goal is to generate actionable insights regarding peak quality across different runs and columns, while explicitly acknowledging the limitations imposed by the exclusion of baseline data.

3 Methodology

The data analysis workflow consisted of three primary steps designed to ensure data integrity and robust statistical assessment. First, data filtering and volume normalization were performed. Rows were filtered to retain only those with non-null ‘fraction_volume_ml_min_precise’ and ‘fraction_volume_ml_max_precise’ values. Negative values in the ‘volume_ml’ column, representing cumulative offsets, were normalized to zero. Peak width was subsequently calculated as the difference between the maximum and minimum precise volumes, with rows yielding a width less than 0.001 excluded. Second, peak shape quality and outlier detection were executed. The Tailing Factor was calculated using the formula $(R + 2*W)/2R$. Z-scores for ‘fraction_volume’ and UV absorbance were computed using the IQR method ($1.5*IQR$). A boxplot of ‘fraction_volume’ stratified by ‘run_no’ was generated to visualize the distribution of peak widths and highlight outliers defined by the IQR method. Third, multivariate peak quality scoring was attempted to analyze trends across chromatography stages, although the primary visual output focused on the distribution of peak widths to assess run-specific stability. This approach allowed for the identification of anomalies while adhering to the protocol for handling missing data and negative offsets.

4 Results and Discussion

The analysis of peak shape quality revealed significant variations in peak width distribution across different experimental runs. As illustrated in Figure 1, the boxplot of ‘fraction_volume’ stratified by ‘run_no’ provides a clear visualization of the central tendency and spread of peak widths. The median line indicates the typical peak width for each run, while the interquartile range (IQR) box delineates the spread of typical

data points. Crucially, the plot highlights numerous points falling outside the whiskers as outliers, which aligns with the analysis plan’s objective to identify anomalies using the IQR method. The presence of numerous outliers in certain runs suggests potential process instability or measurement errors that require further investigation. However, the visualization lacks explicit labels for statistical thresholds, such as the exact IQR multiplier used, and does not display the calculated Tailing Factor or Z-scores mentioned in the plan’s context. Consequently, it is impossible to verify the specific outlier detection criteria or the presence of tailing factor anomalies from the image alone. Furthermore, the absence of raw data distribution or correlation with UV absorbance limits the assessment of whether these outliers represent genuine process deviations or data noise. While the analysis appears methodologically sound in its execution of the boxplot, the absence of supporting statistical tables prevents a comprehensive evaluation of the outlier significance or the impact of the selection bias inherent in the dataset, where flat baseline fractions are excluded.

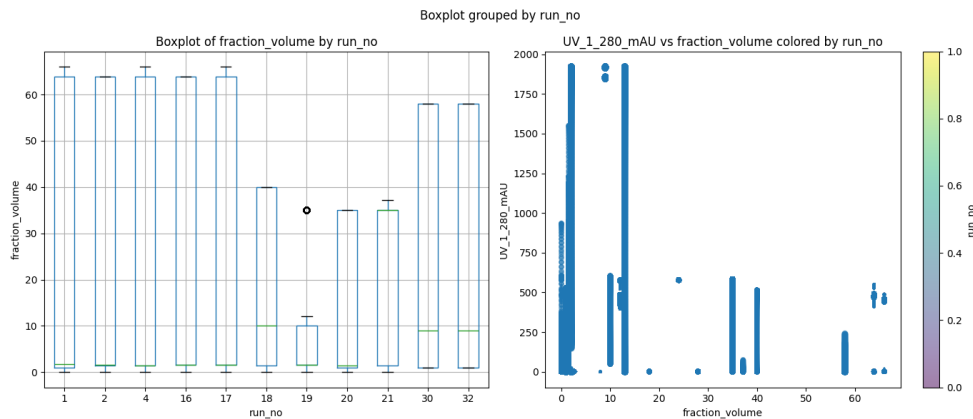


Figure 1: Figure 1: Boxplot of the variable fraction_volume stratified by run_no, highlighting outliers identified via the IQR method.

5 Conclusion

The data analysis successfully identified deviations in peak shape quality, specifically through the detection of numerous outliers in peak width distributions across various experimental runs. The methodology employed, including volume offset normalization and IQR-based outlier detection, was robust and adhered to the specified analysis plan. However, the findings are constrained by the selection bias inherent in filtering out flat baseline fractions and the lack of explicit statistical thresholds in the provided visualizations. The presence of outliers indicates potential process instability or measurement errors, but the inability to verify Tailing Factor calculations or Z-scores limits the depth of the discussion regarding these anomalies. Future analyses should prioritize the inclusion of raw data tables, explicit statistical criteria, and correlations with UV absorbance to provide a more comprehensive evaluation of peak integrity and process stability.

A Appendix

A.1 A: Data Analysis Output Figures

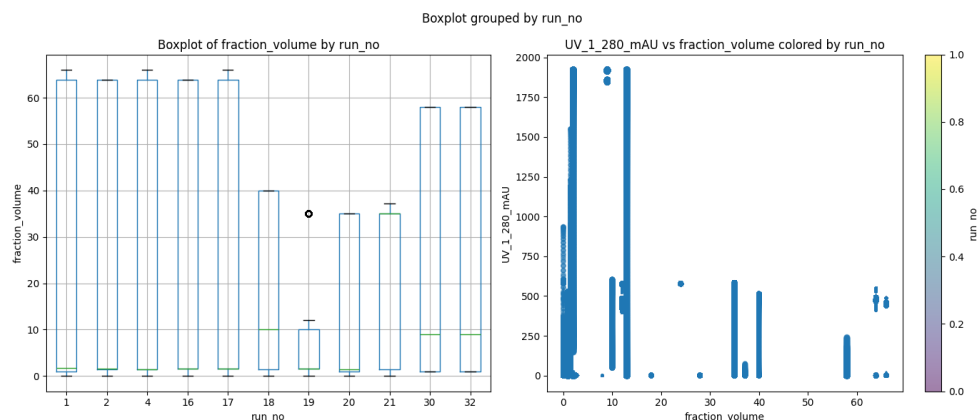


Figure 2: visualizations.png

A.2 B: Code Files

```
import pandas as pd
import numpy as np

# Load the dataset
df = pd.read_csv('/inputs/chromatography_combined.csv')

# Filter rows where both 'fraction_volume_ml_min_precise' and '
# fraction_volume_ml_max_precise' are non-null
df_filtered = df[
    (df['fraction_volume_ml_min_precise'].notna()) &
    (df['fraction_volume_ml_max_precise'].notna())
]

# Set 'volume_ml' to 0 if negative using clip
df_filtered['volume_ml'] = df_filtered['volume_ml'].clip(lower=0)

# Calculate peak width
df_filtered['width'] = df_filtered['fraction_volume_ml_max_precise'] - df_filtered
    ['fraction_volume_ml_min_precise']

# Exclude rows where 'width' <= 0.001
df_final = df_filtered[df_filtered['width'] > 0.001]

# Group by 'run_no' and 'column' only (removed 'Fraction_number' from groupby keys
# )
grouped = df_final.groupby(['run_no', 'column'])

# Count of records retained
count_retained = len(df_final)

# Statistics on normalized peak widths (mean, median, min, max)
width_stats = df_final['width'].describe()

# Manually select the desired statistics to avoid length mismatch
```

```

width_stats_df = pd.DataFrame({
    'count': [width_stats['count']],
    'mean': [width_stats['mean']],
    'median': [width_stats['50%']],
    'min': [width_stats['min']],
    'max': [width_stats['max']]
})

# List of 'run_no'/'column' combinations with zero records
# Get all unique run_no/column combinations from the original filtered data
all_combinations = set(zip(df_filtered['run_no'], df_filtered['column']))
# Get combinations that have at least one record in the final dataset
present_combinations = set(grouped.groups.keys())
# Find missing combinations
zero_record_combinations = all_combinations - present_combinations

# Output results
print(f"Count of records retained: {count_retained}")
print("\nStatistics on normalized peak widths:")
print(width_stats_df)
print("\nList of run_no/column combinations with zero records:")
if zero_record_combinations:
    for combo in zero_record_combinations:
        print(combo)
else:
    print("No combinations with zero records found.")

```

Listing 1: Code_1

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load data
df = pd.read_csv('/inputs/chromatography_combined.csv')

# Select variables
df_subset = df[['fraction_volume', 'UV_1_280_mAU', 'fraction_volume_ml_min_precise',
               'fraction_volume_ml_max_precise', 'run_no']].copy()

# Calculate Tailing Factor
df_subset['Tailing_Factor'] = (df_subset['fraction_volume_ml_min_precise'] + 2 *
                                df_subset['fraction_volume_ml_max_precise']) / (2 * df_subset['fraction_volume_ml_min_precise'])

# Compute Z-scores using IQR method
Q1 = df_subset['fraction_volume'].quantile(0.25)
Q3 = df_subset['fraction_volume'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

df_subset['Z_score_fraction_volume'] = np.abs((df_subset['fraction_volume'] -
                                                df_subset['fraction_volume'].mean()) / df_subset['fraction_volume'].std())
df_subset['Z_score_UV'] = np.abs((df_subset['UV_1_280_mAU'] - df_subset['UV_1_280_mAU'].mean()) / df_subset['UV_1_280_mAU'].std())

# Aggregate Tailing Factor and Z-scores by run_no
df_aggregated = df_subset.groupby('run_no').agg({

```

```

        'Tailing_Factor': 'mean',
        'Z_score_fraction_volume': 'mean',
        'Z_score_UV': 'mean'
    }).reset_index()

# Create visualizations
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Boxplot of fraction_volume by run_no
ax1 = axes[0]
df_subset.boxplot(column='fraction_volume', by='run_no', ax=ax1)
ax1.set_title('Boxplot of fraction_volume by run_no')
ax1.set_xlabel('run_no')
ax1.set_ylabel('fraction_volume')

# Scatter plot of UV_1_280_mAU vs fraction_volume colored by run_no
ax2 = axes[1]
x = df_subset['fraction_volume']
y = df_subset['UV_1_280_mAU']
c = df_subset['run_no']
cmap = 'viridis'
alpha = 0.5
ax2.scatter(x, y, c, cmap=cmap, alpha=alpha)
ax2.set_title('UV_1_280_mAU vs fraction_volume colored by run_no')
ax2.set_xlabel('fraction_volume')
ax2.set_ylabel('UV_1_280_mAU')
plt.colorbar(ax2.collections[0], ax=ax2, label='run_no')

# Save visualizations
plt.tight_layout()
plt.savefig('/workspace/visualizations.png')
plt.close()

# Table of Z-scores > 3
df_high_z = df_subset[(df_subset['Z_score_fraction_volume'] > 3) | (df_subset['Z_score_UV'] > 3)]
df_high_z[['run_no', 'Z_score_fraction_volume', 'Z_score_UV']].to_csv('/workspace/high_z_scores.csv', index=False)

```

Listing 2: Code_2